Batterjee Medical College
For Science and Technology

Hematological Disorders

Internal Medicine Department

Learning Objectives

Knowledge

- Relate pathophysiological basics and etiology of anaemia
- Describe epidemiology, manifestations, complications and management of anaemia

Skills

- Interpret clinical manifestations and investigations of anaemia to formulate reasonable diagnosis.
- Apply evidence-based management plan for anaemia
- Distinguish patients in emergency situations to formulate a comprehensive management plan; ensuring keeping patients in normal homeostasis

Main Subtopics

- Blood Components & Reference Ranges (CBC parameters, ESR)
- Hematopoiesis Sites, Stages & Erythropoiesis Requirements
- RBC Morphology & Inclusions (Anisocytosis, Poikilocytosis, Howell-Jolly bodies)
- Iron Deficiency Anemia (Pathogenesis, Iron panel interpretation, Treatment)
- Differential Diagnosis of Microcytic Anemias (IDA, ACD, Thalassemia, Sideroblastic)

Blood overview

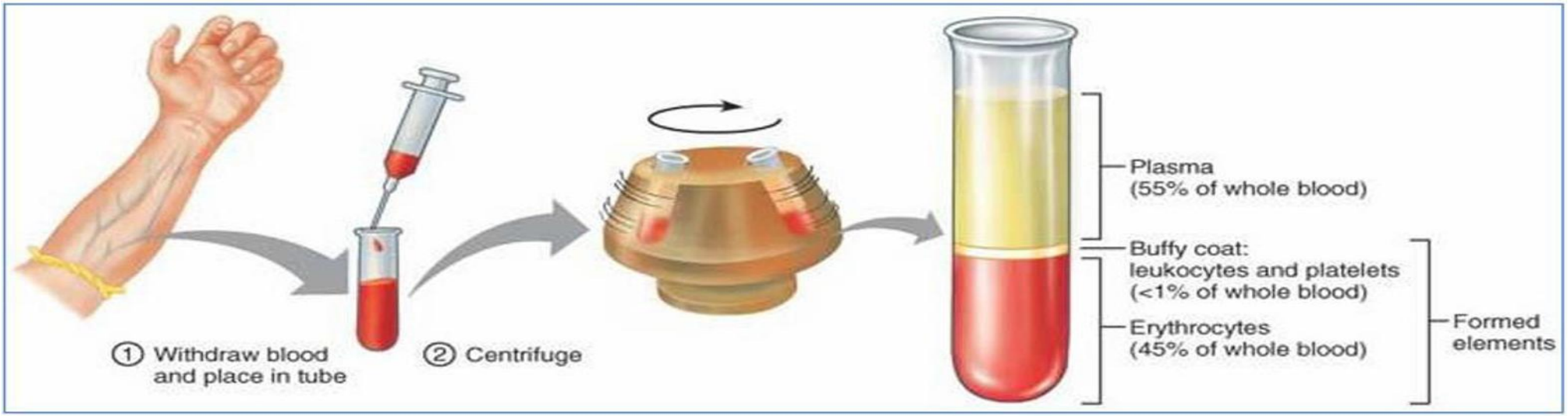
The blood is composed of plasma and 3 cell types:

Plasma: Straw-coloured, sticky fluid.

- Mostly water (90%). With dissolved nutrients, hormones, ions(Na, K, …) and protein.
- **5-7% Protein:**
  - **Albumin → blood carrier,**
  - **Globulin → mainly immunoglobulins, Fibrinogen → part of a clotting protein**

Serum: The fluid, noncellular portion of blood that remains after coagulation. The serum is equivalent to plasma without its clotting elements.

Cells: RBCs, WBCs, Platelets.



Components of the Complete Blood Count (CBC) Report

- **<u>1- RBCs Parameters:</u>**
- Haemoglobin (**Hb**).
- Red blood cell count (**RBCs**).
- Haematocrit (**Hct**).
- Mean corpuscular volume (**MCV**).
- Mean corpuscular Haemoglobin (**MCH**).
- Mean corpuscular Haemoglobin Concentration (**MCHC**).
- Reticulocytic count.
- Red cell distribution width (**RDW**).
- **<u>2- Platelets count (PLT).</u>**
- **<u>3- WBCs parameters:</u>**
- White blood cell count (**WBCs**)
- White blood cell differential (Relative & Absolute Count).

Haematology

CBC-Complete Blood Count

Parameters	Result(s)	Unit(s)	Reference Value(s)
Total WBC	6.55	x10^9/L	4.0-11.0
RBC	**6.24**	x10^12/L	M:4.5-6.0 F:4.0-5.0
Hemoglobin.	17.2	g/dl	M:14-18 F:12-16
HCT	50.1	%	M:40-54 F:37-47
MCV	80.3	fL	75-96
MCH	27.6	pg	27-32
MCHC	34.3	g/dl	31-35
Platelet Count	210	x10^9/L	150-450
Neutrophils	57.5	%	40 - 75
Lymphocytes	32.3	%	20 - 40
Monocytes	7.8	%	2 - 10
Basophils	0.0	%	0 - 1
Eosinophils	2.4	%	1 - 6
MPV	13.2	fL	6.8 - 10.2

Haematopoiesis (Formation of ALL Blood Cells)**:** 'The Process of development, differentiation, and maturation of All Blood Cells'. All blood cells are derived from pluripotent stem cells.

Erythropoiesis is regulated by the Hormone 'Erythropoietin'.

Requirements for Erythropoiesis & Haemoglobin Formation:

- The *Marrow* requires other precursors for effective erythropoiesis: eg.
  - **Metals:**
    - Iron – essential for Haemoglobin synthesis
  - **Vitamins:**
    - Especially Vit. B_{12} - necessary for normal DNA synthesis
    - Folate - necessary for normal DNA synthesis
  - **Amino Acids:**
    - For the production of proteins
  - **Hormones:**
    - ***Erythropoietin***

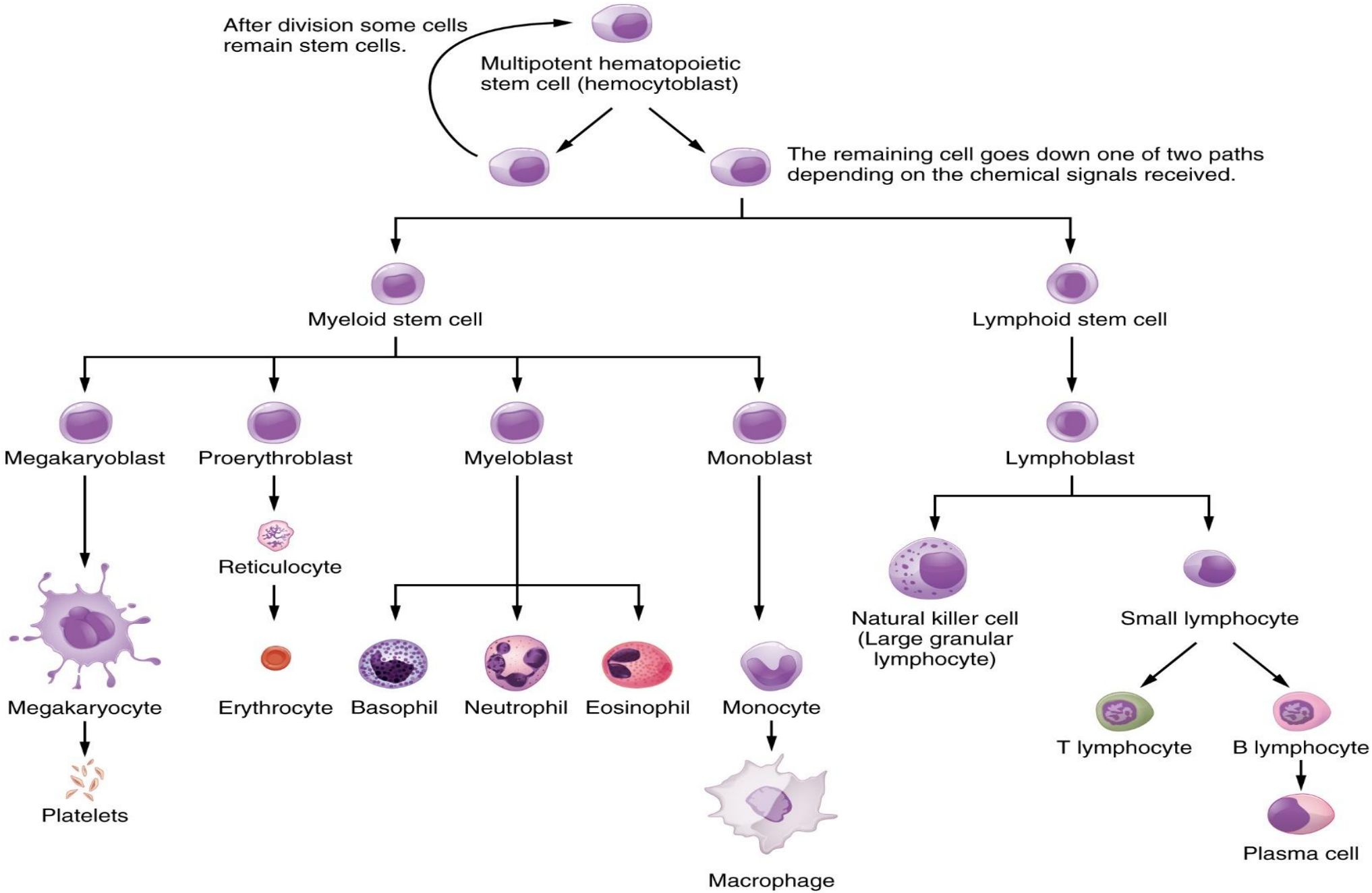
Erythrocytes:

- Foetal Life: Takes place in the Liver/Spleen/&Bone Marrow.
- Childhood& Adult: Occurs only in the bone marrow (**Medullary cavity**),i.e., The Bone Marrow is the only source of new blood cells.
- In some diseases (at times of extreme demand. e.g. During Anemia), Hematopoiesis occurs in the liver & spleen (**Extramedullary erythropoiesis**)

Leukocytes:

- Granulocytes and monocytes are formed in the bone marrow from common precursor cells (Myeloid lineage).

Lymphocytes: are formed in the bone marrow and thymus (Primary Lymphoid Organs) and spleen, lymph nodes, tonsils, adenoids, and payer's patches (Secondary Lymphoid Organs).


After division some cells remain stem cells.
Multipotent hematopoietic stem cell (hemocytoblast)
The remaining cell goes down one of two paths depending on the chemical signals received.
Myeloid stem cell
Lymphoid stem cell
Megakaryoblast
Proerythroblast
Myeloblast
Monoblast
Lymphoblast
Reticulocyte
Megakaryocyte
Erythrocyte
Basophil
Neutrophil
Eosinophil
Monocyte
Natural killer cell (Large granular lymphocyte)
Small lymphocyte
T lymphocyte
B lymphocyte
Plasma cell
Platelets
Macrophage

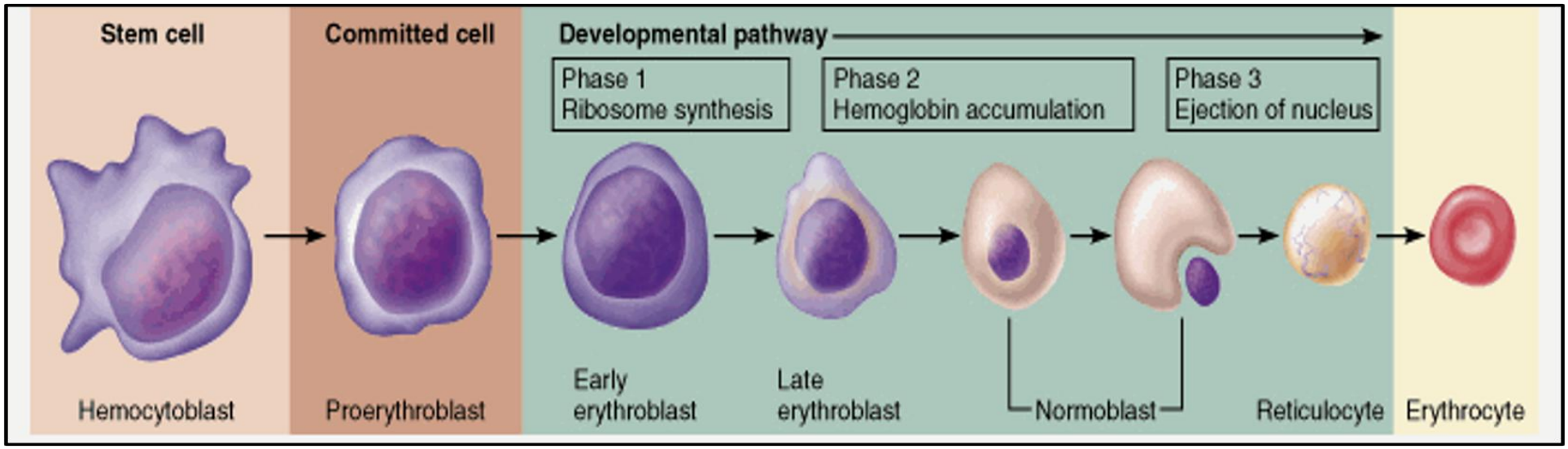
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Red Blood Cells

- **Yolk sac** in the third week of intrauterine life.
- **Liver and spleen** by the 3rd month.
- 4th month of intrauterine life: **BM** starts to contribute
- In adult life, it is restricted to the **ends of long bones** (head of femur) and **Flat bones** (skull, sternum, pelvis, and ribs).

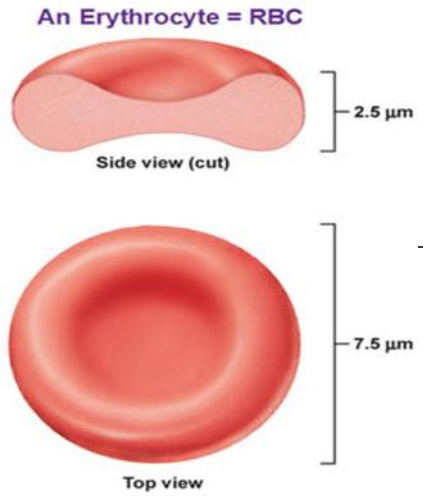
Stages of development

- Totipotent stem cells → pleuripotent stem cells→ stem cells committed for erythropoiesis→ Proerythroblasts→ erythroblasts→ normoblasts→ reticulocytes (containing reticulum of residual RNA) → RBCS

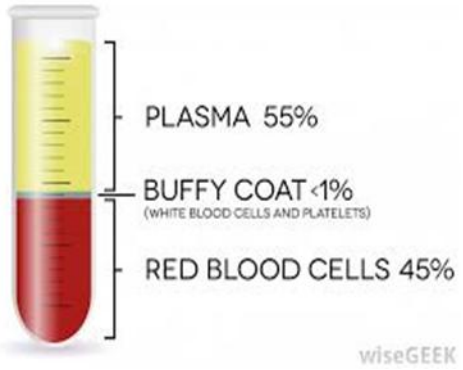


Hematological Indices

(1) Shape	(2) Count	(3) Diameter
Normal: Biconcave **Abnormal**: - Spherocytes - Sickle cells - Target cells	**Normal**: ♂: 4.5 – 6 million/mm3 ♀: 3.8-4.8 million/mm3 **Abnormal**: Count = anemia Count = polycythemia	**Normal** = 7.2 micron **Abnormal**: < 6.7 → microcyte > 8 → Macrocyte > 12 → Megalocyte Variable →anisocytosis



(4) Hematocrite (Packed cell volume) = PCV or HCT



- Volume of packed RBCs in 100 ml blood
- Normal = 45%:
  - ↑ PCV = Dehydration or polycythemia.
  - ↓ PCV = Hypervolemia or anemia.

(5) Mean corpuscular volume = (volume of RBC): (78-98 FL)

$$\frac{\text{Hematocrite (volume of RBCs) 45 ml}}{\text{RBC count}} = \frac{45\text{ml}}{5\text{ millon}} = 90u^3$$

- ↑MCV (> 100) = Macrocytic
- ↓ MCV (< 78) = Microcytic

(6) Hemoglobin:

Normal: ♂: 13.5 – 16 gm%. ♀: 11.5 – 13.5 gm%.

- ↓ Anemia or hypervolemia.
- ↑ Polycythemia or dehydration.

(7) Mean corpuscular hemoglobin concentration (MCHC):

$$= \frac{\text{Hb / 100 ml blood}}{\text{Hematocrite}} \times 100 = 32 – 36\%$$

(8) Mean corpuscular hemoglobin:

$$= \frac{\text{Hb / 100 ml blood}}{\text{RBCs count/mm}} \times 10 = 30 \text{ Pg}$$

(9) Erythrocyte sedimentation rate (ESR):

- Normally, RBCs carry negative charges→ repulsion.
- Negative charges on RBCs: Albumin ↑ & fibrinogen & globulin ↓
- -ve charges prevents RBCs to from roulex & to sediment.
- Vertical distance sedimented by RBCs in blood column at end of 1st & 2nd hr is known ESR.
- It's measured by Westergnen's method (Blood +Na citrate).

	1st hour	2nd hour	
· **Normally:**	5 mm	8 mm	Male
	10 mm	16 mm	Female

Cause of ↑ ESR	Cause of ↓ ESR
○ ↑ Fibrinogen: In pregnancy & menses. ○ ↑ Globulin: ▪ Multiple myeloma. ▪ Collagen diseases. ▪ Malignancy ▪ Infection. ○ ↓ Albumin: Nephrotic syndrome	○ Polycythemia. ○ Afibrinogenemia. ○ Heart failure.

o**Value:** ESR is used to follow up disease progress & response to treatment not for diagnosis

○ **Very high ESR(> 100 mm/HR):**

1. Malignancy
2. Collagen diseases
3. Active TB

○ **Anemia with low ESR**: 1. Sickle cell anaemia 2. Hereditary spherocytosis

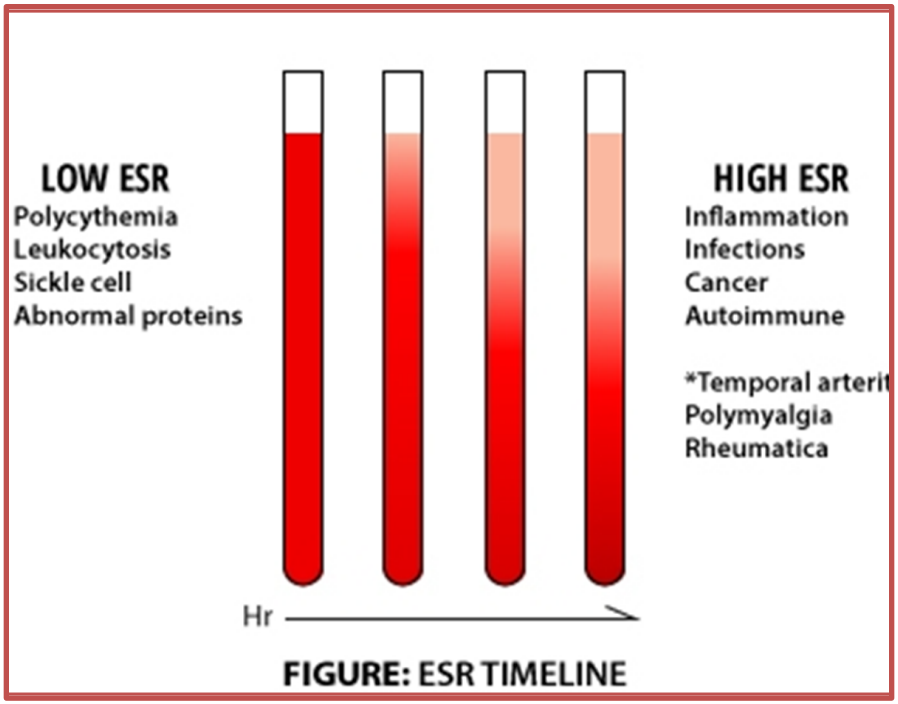


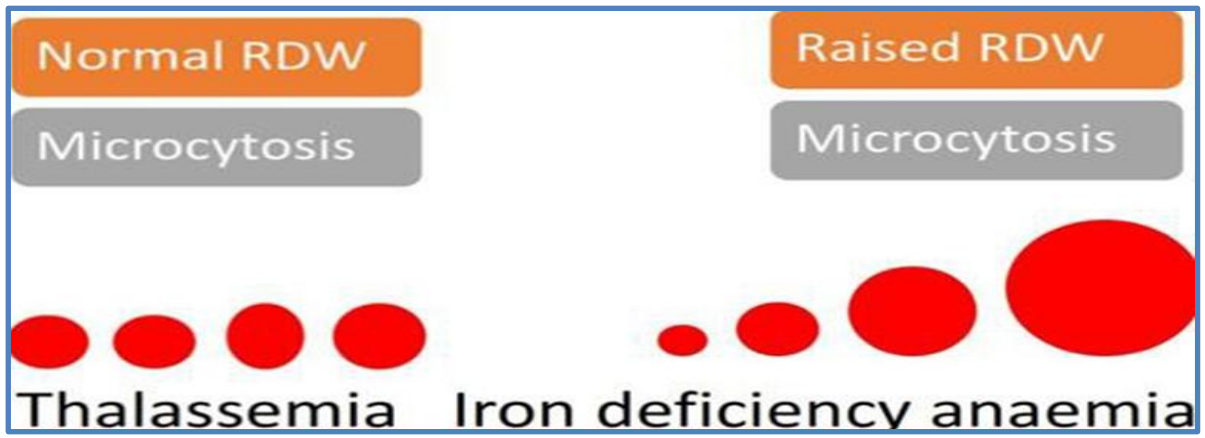
FIGURE: ESR TIMELINE

(10) Reticulocytes: Value: indicator of BM activity

Normally: 0.2-2% of RBCs are reticulocytes.

Reticulocytosis	Absent
• Hemolysis • Acute hemorrhage • Anemia with treatment	• BM failure

Red blood cell distribution width (RDW or RDW-CV or RCDW and RDW-SD) is a measure of the range of variation of red blood cell



Polychromasia is a disorder where there is an abnormally high number of immature red blood cells found in the bloodstream as a result of being prematurely released from the bone marrow during blood formation.

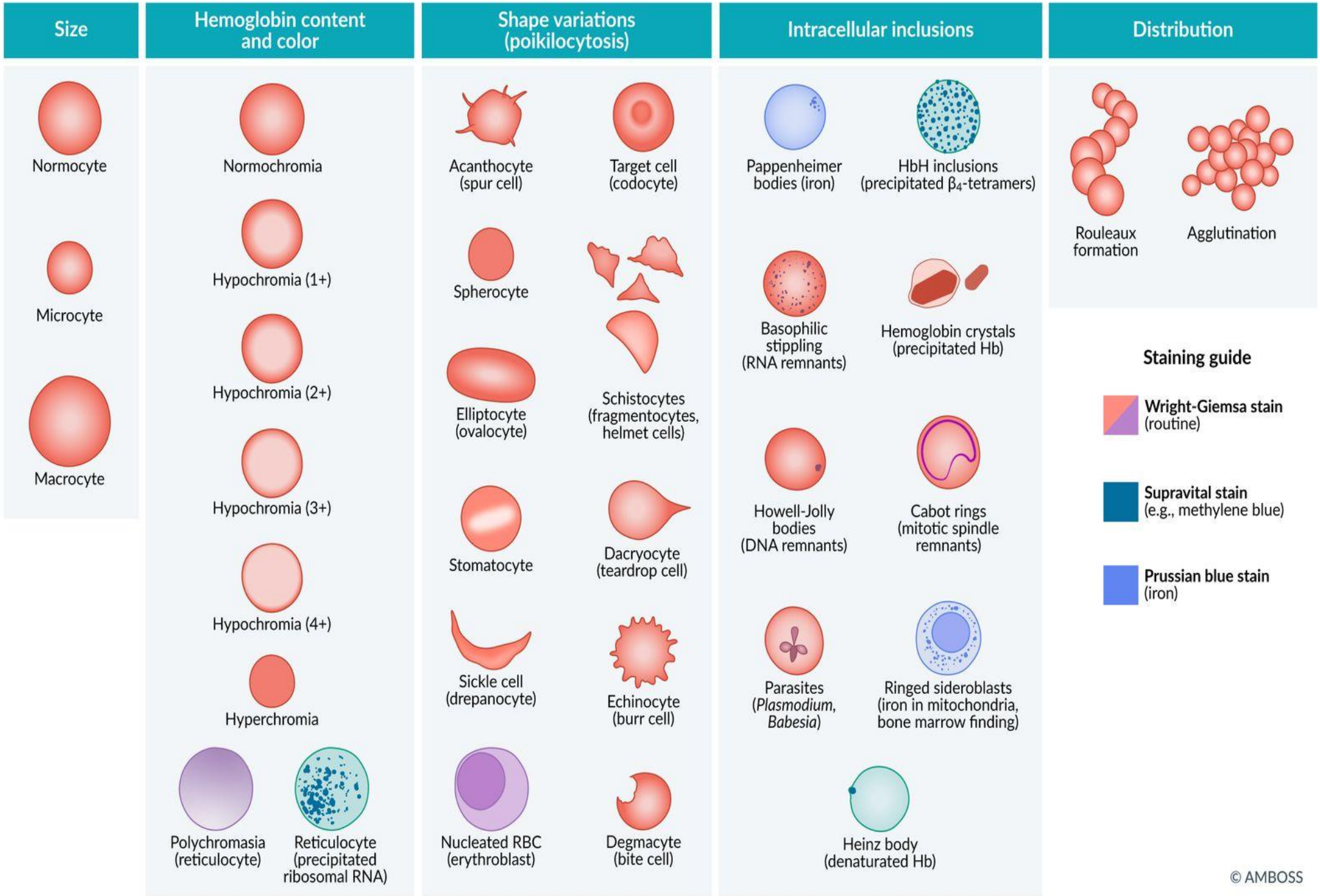
Value of blood film

1. **Abnormal RBCs shape:**

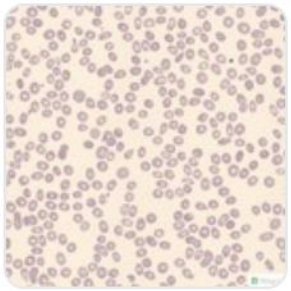
2.

- **Anicocytosis (variation in size):** Megaloblastic or iron deficiency (non specific), associated with increased red cell distribution width
- **Poikilocytosis:** (Vriaion in shape): non specific
- **Punctate basophilia:** (basophilic stippling) lead poisoning& thallassemia.
- **Polychromasia:** causes of increased reticulocytes
- **Parasites in RBCs:** malaria
- **Howell jolly bodies:** Megaloblastic anemia, post splenecomy.

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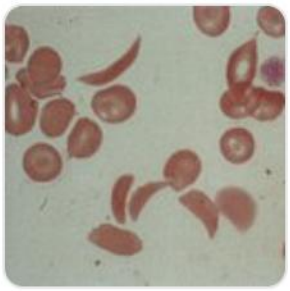
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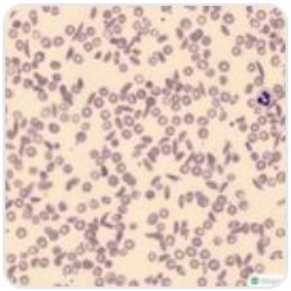
Dacrocytes



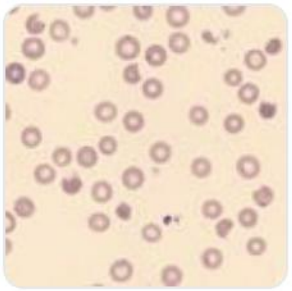
Dacryocytes



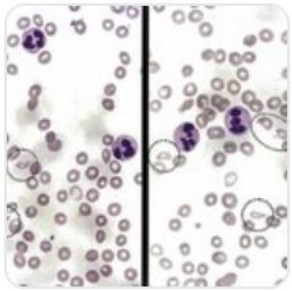
Sickle cell disease with drepanocytes and target cells



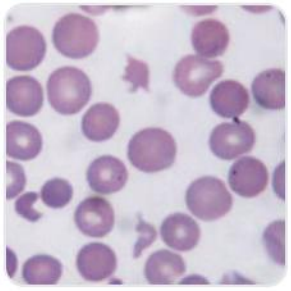
Erythrocyte morphologies in sickle-cell disease



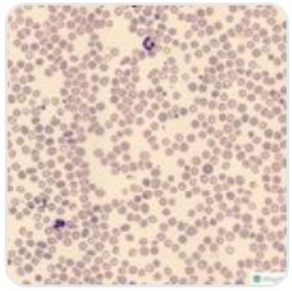
Schistocytes



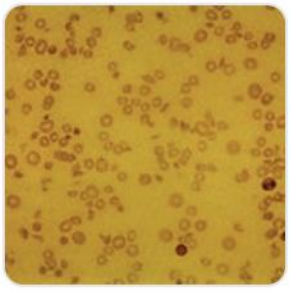
Schistocytes in hemolytic uremic syndrome



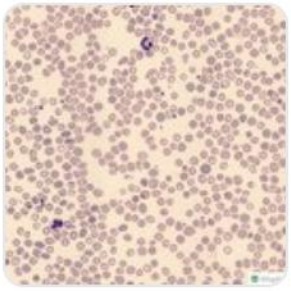
Schistocytes



Macrocytes



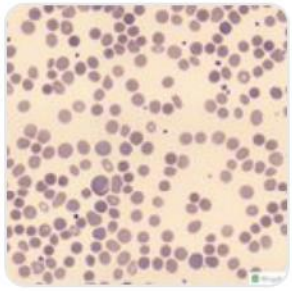
Schistocytes in hemolytic uremic syndrome



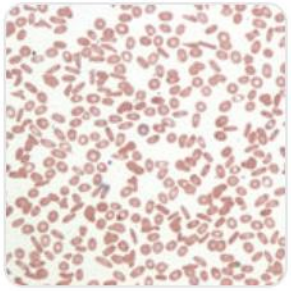
Macrocytes



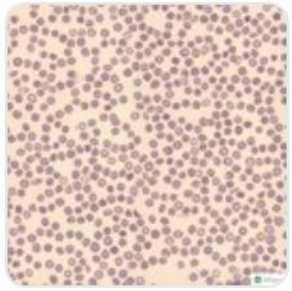
Macroovalocyte



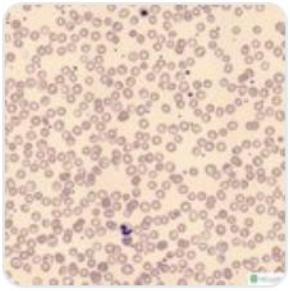
Spherocytosis



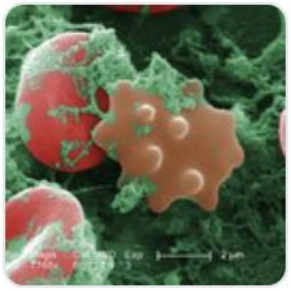
Hereditary elliptocytosis



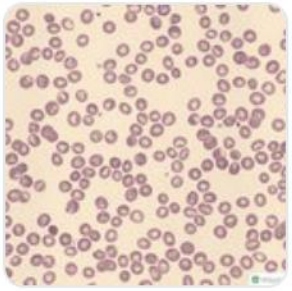
Echinocytes



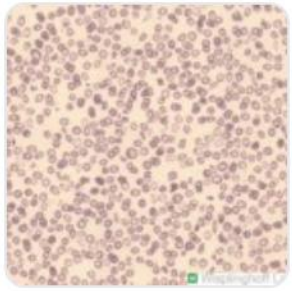
Target cells (codocytes)



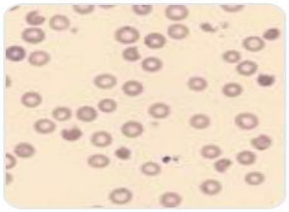
Acanthocyte



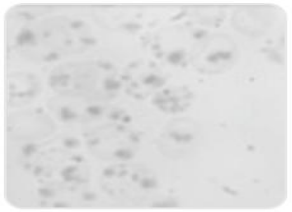
Stomatocytes



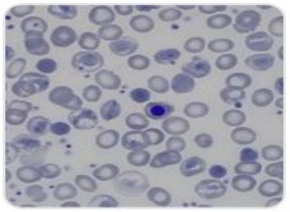
Anisocytosis and poikilocytosis



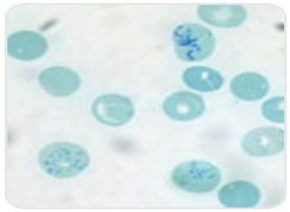
Schistocytes



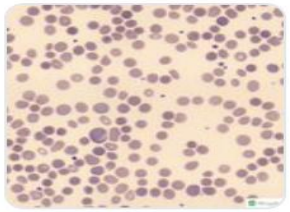
Heinz bodies



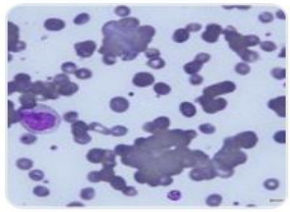
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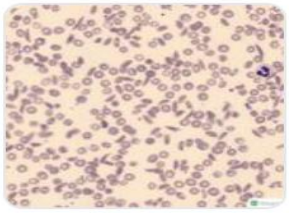
Reticulocytes in peripheral blood



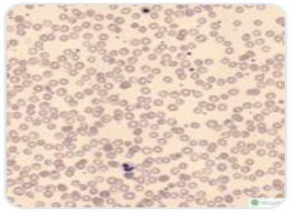
Spherocytosis



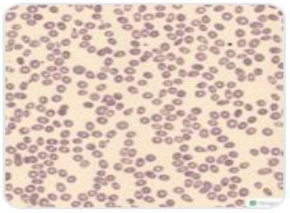
Agglutination of erythrocytes



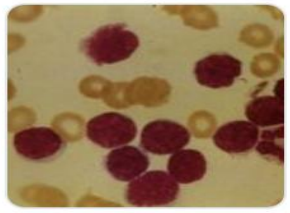
Erythrocyte morphologies in sickle-cell disease



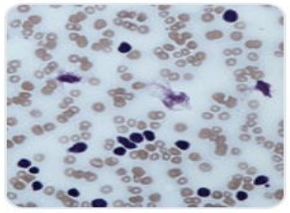
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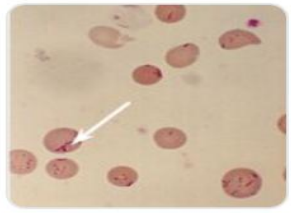
Dacrocytes



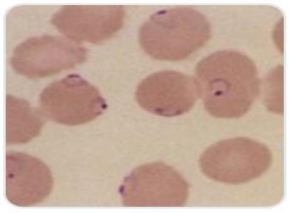
Smudge cells in chronic lymphocytic leukemia (CLL)



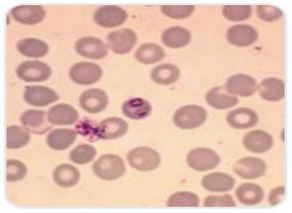
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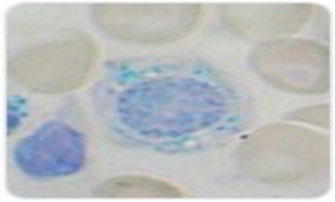
Bartonella bacilliformis infection (Carrión...



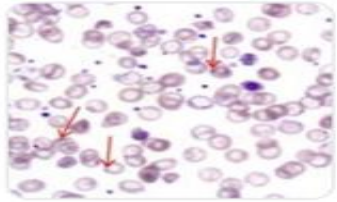
Falciparum malaria: immature trophozoite in...



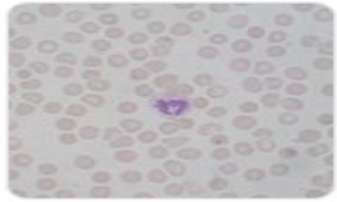
Babesia microti (blood smear)



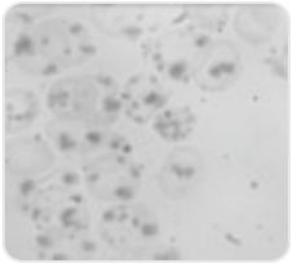
Ringed sideroblast



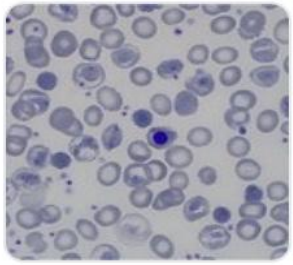
Howell-Jolly bodies in asplenia



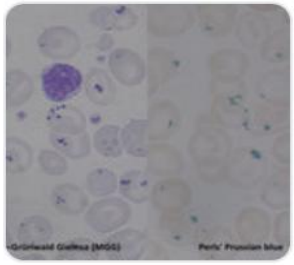
Howell-Jolly bodies



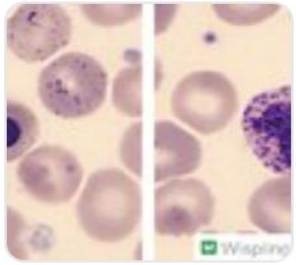
Heinz bodies



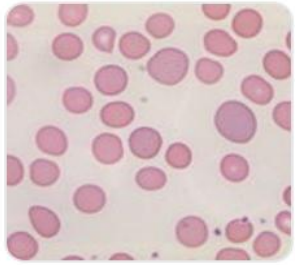
Heinz bodies



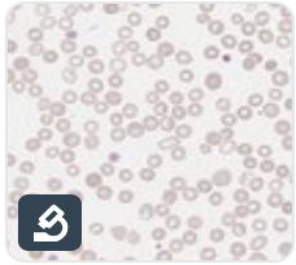
Pappenheimer bodies



Basophilic stippling of erythrocytes



Basophilic stippling of erythrocytes



Basophilic stippling of erythrocytes

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Anemias

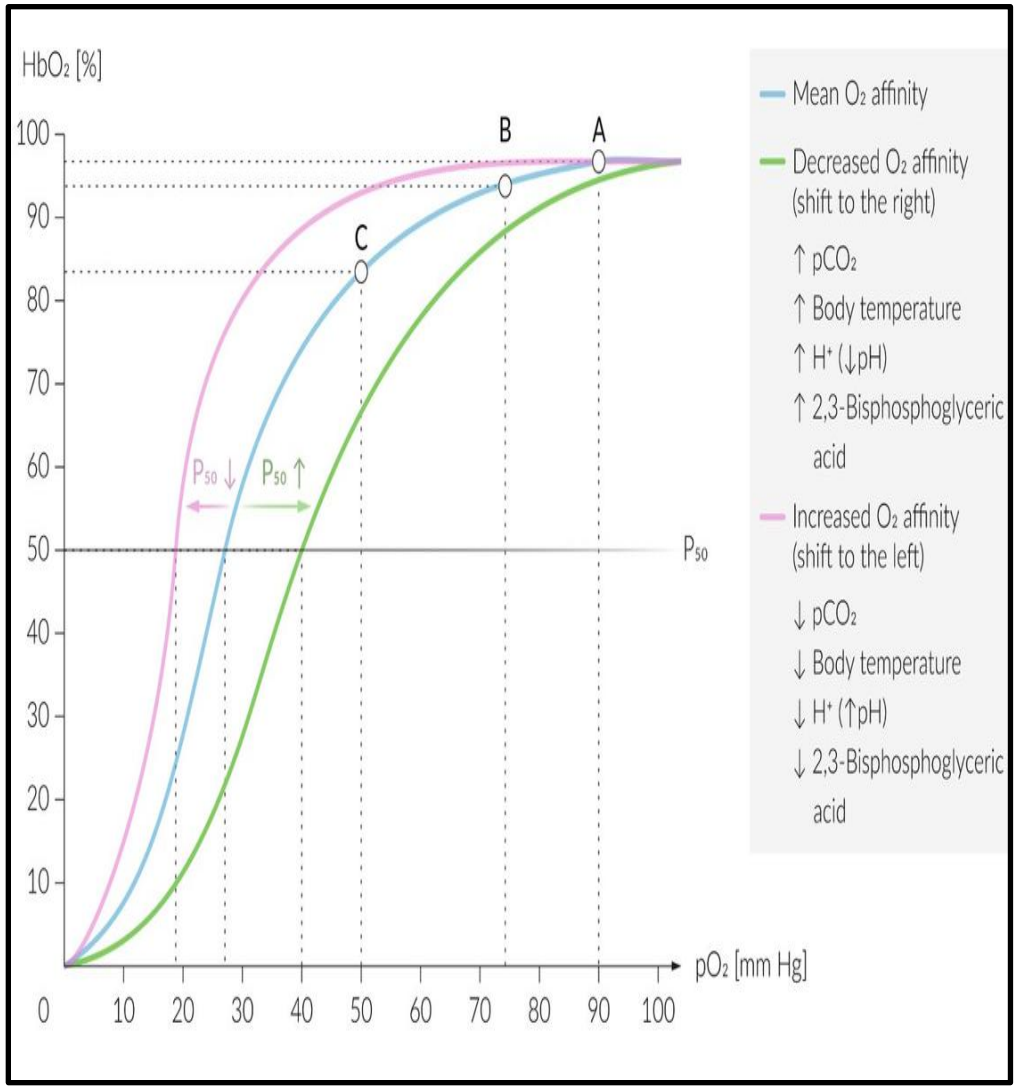
Definition / Etiology

Definition

- Reduced O2 carrying capacity of blood.
- It's identified as Hb < 12 gm% in Females or Hb < 13 gm% in males.

Pathophysiology:

1. Hyperdynamic circulation → Short circulation time & high cardiac output.
2. ↑ Erythropoietin by the kidney→ +++ RBCs
3. ↑O2 delivery to the tissue (shift of O2 dissociation curve to the right): This is due to excess production of 2,3-DPG (2,3-diphosphoglycerate), which makes the binding between O2 & Hb very weak.
4. Redistribution of the blood: blood is shifted to the vital organs e.g., the brain.



Clinical picture

A. General picture of anemia



1. Pallor.

2. General systems:

a. CNS:

- Dizziness, Tinnitus, lack of concentration & blurred vision.
- Tingling & numbness (PN).

- b. CVS: Hyperdynamic circulation:
  - Tachycardia & palpitation, capillary pulsations, & increased pulse pressure(water hummer pulse).
  - Functional murmurs.
  - Anemic heart failure.



c. GIT: Dyspepsia – anorexia.

d. Renal: Polyuria, proteinuria (Hypoxia→VD of renal BVs).

e. Reproductive organs: amenorrhea or menorrhagia.

3. Oedema: due to:

- Hypoxia →↑ capillary permeability.
- Hypoproteinemia (associated nutritional deficiency) →↓ osomotic activity of blood.
- Anemic heart failure.

B. Specific features for each anemia.

C. C/P of cause.

Investigations

A. As in all types of anemia: ↓ RBCs, PCV, ↑ ESR

B. Type of anemia:

Microcytic hypochromic:	**Normocytic normochromic:**	**Macrocytic normochromic:**
• ↓ MCV. • ↓ MCH. • ↓ MCHC	• MCV; normal. • MCH.: normal • MCHC: normal	• ↑ MCV. • MCH.: normal • MCHC: normal

C. Investigation of the cause of anemia.

D. Investigation of etiology.

Treatment

1. Treatment of the cause.

2. Blood transfusion if:

- Hb < 7 gm% or
- there is HF or
- ischemic heart disease.

Classification of anemia

I. According to etiology:

Deficiency anemia - bone marrow failure – Hemolytic – Hemorrhagic

II. According to type of anemia:

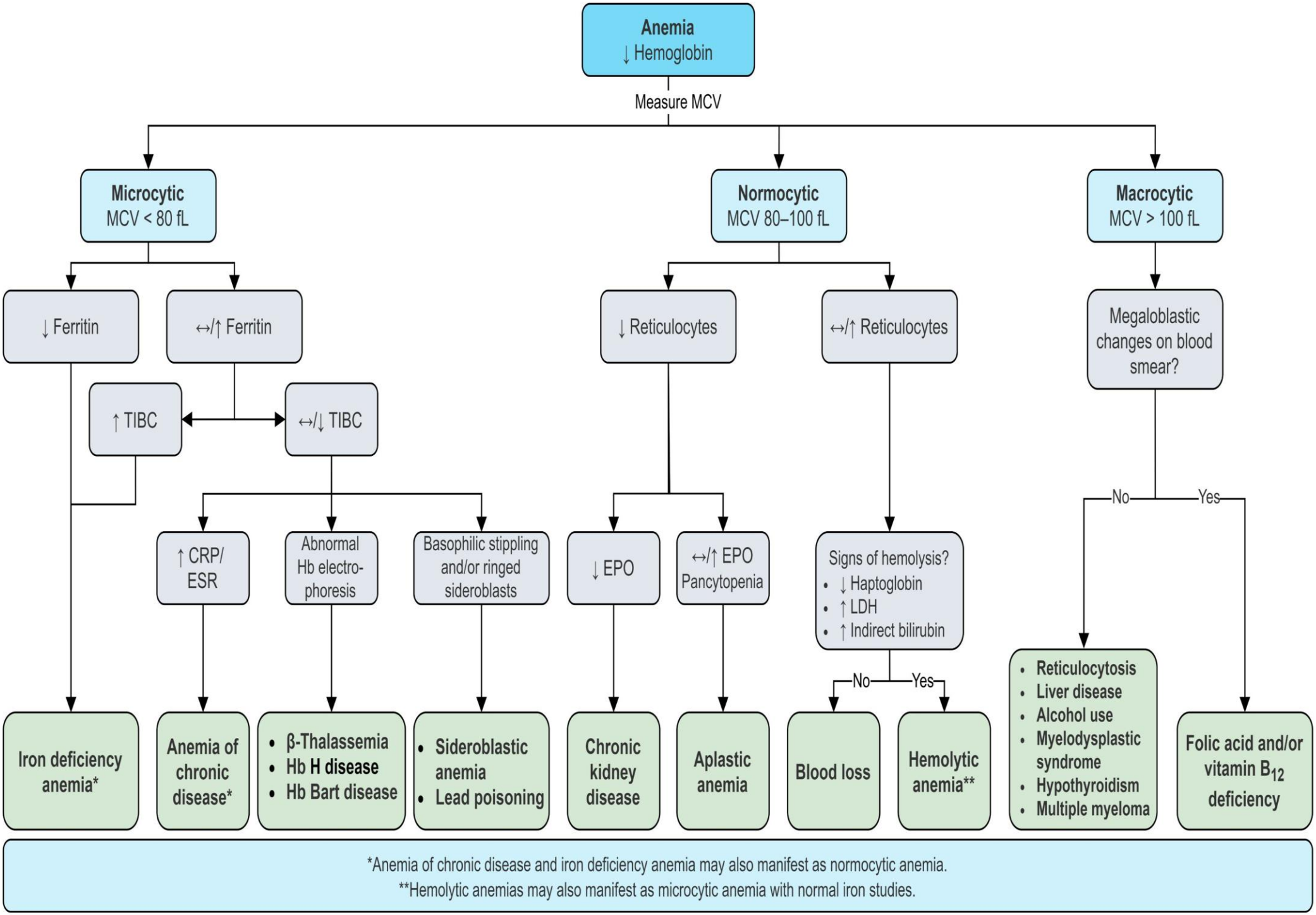
- ***Normocytic normochromic anemia:***
  1. Anemia of BM failure
  2. Acute hemorrhage
  3. Hemolysis

- ***Microcytic hypochromic anemia***
  1. Iron deficiency anemia
  2. Thallassemia
  3. Sideroblastic anemia
  4. Anemia of chronic illness
  5. Chronic lead intoxication

- ***Macrocytic normochromic anemia:***

  I. Megaloblastic: Folic acid or Vit B12 anemia

  II. Non megaloblastic:
  - Liver cell failure
  - Hypothyroidism
  - Myelodysplasia.
  - Alcoholism


Anemia
↓ Hemoglobin
Measure MCV
Microcytic
MCV < 80 fL
Normocytic
MCV 80–100 fL
Macrocytic
MCV > 100 fL
↓ Ferritin
↔/↑ Ferritin
↑ TIBC
↔/↓ TIBC
↓ Reticulocytes
↔/↑ Reticulocytes
Megaloblastic changes on blood smear?
↑ CRP/ ESR
Abnormal Hb electro-phoresis
Basophilic stippling and/or ringed sideroblasts
↓ EPO
↔/↑ EPO Pancytopenia
Signs of hemolysis?
• ↓ Haptoglobin
• ↑ LDH
• ↑ Indirect bilirubin
No
Yes
No
Yes
Iron deficiency anemia*
Anemia of chronic disease*
• β-Thalassemia
• Hb H disease
• Hb Bart disease
• Sideroblastic anemia
• Lead poisoning
Chronic kidney disease
Aplastic anemia
Blood loss
Hemolytic anemia**
• Reticulocytosis
• Liver disease
• Alcohol use
• Myelodysplastic syndrome
• Hypothyroidism
• Multiple myeloma
Folic acid and/or vitamin B12 deficiency
*Anemia of chronic disease and iron deficiency anemia may also manifest as normocytic anemia.
**Hemolytic anemias may also manifest as microcytic anemia with normal iron studies.

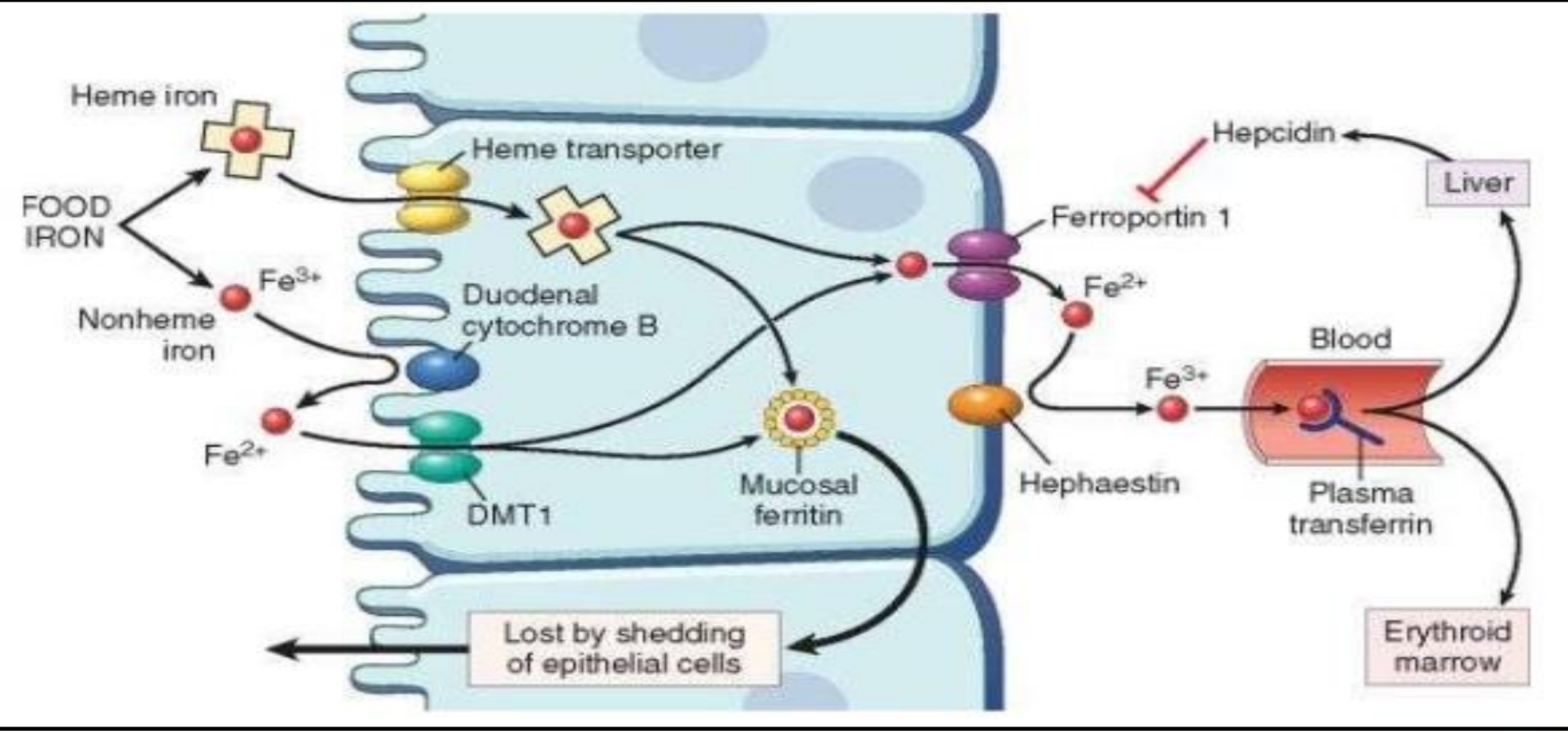
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Iron Deficiency Anemia

Physiology

- **Source:** in meat, liver, spinach.
- **Requirements:** 10 mg /day (only 1 mg is absorbed)
- **Absorption:** iron is converted from **ferric to ferrous** state in the stomach by HCL. Absorption in the duodenum is facilitated by **apoferrtin** transport system, which is saturable system that allows only 10 % of iron intake to be absorbed **(Mucosal block theory)**.
- **Functions:**
  - Myoglobin & Hb synthesis.
  - Synthesis of heme containing enzymes e.g. cytochrome oxidase and catalaze.
- **Storage:** 1 gm stored as ferritin or hemosidrin in RES.

Some ferritin is circulating (♀ 10-200 ng/ml , ♂15-400 ng/ml). Each 1 ug/L of circulating ferritin is equivalent to 10 mg stored iron.



- ***Transport system:*** iron is transported in the blood bound to transferrin, which is only 30% saturated by iron. Unbound transferrin is called total iron binding capacity (TIBC). Normally transferrin carries 100 Ug/ml of iron & binding capacity is 300 Ug/ml (TIBC = 70%.)

Etiology

I. Diminished intake:

- Starvation, dysphagia, prolonged parenteral feeding without iron supplement
- In children dependent on milk after age of 6 months (mother stores are depleted)

II. Diminished absorption:

- Gastrectomy
- Gastric carcinoma
- Atrophic gastritis
- Malabsorption syndrome
- Excess phytate, phosphate.

III. Increased loss:

- Chronic bleeding e.g. peptic ulcer, esophageal varices, gastritis, cancer colon, piles, heamaturia, hemoptysis, menorrhagia, metrorrhagia.
- Anklystoma doudenale infestation (common cause in Egypt).

IV. Increased requirements:

- Pregnancy, lactation
- Puberty, convalescence

Clinical Picture

I. General *features of anemia:*

II.Specific feature of anemia:

Due to enzymatic defect, cells with high turn over (e.g. epithelial) are the first to be affected:

- Red glazed tongue
- Dyspepsia
- Koilonychia (spooned nails) with loss of luster
- Loss of hair
- Pica (eating unusual substances)

III. C/P of cause e.g.

- Anklystoma infestation: epigastric pain, constipation, parotid enlargement (endemic parotitis), perverted appetite & pellagra (associated malnutrition)

Investigations

1. **↓ Hb, ↓ HCT, ↓ RBCs, ↑ ESR**

2. **Type of anemia:** Microcytic hypochromic: ↓MCV and ↓ MCHC

3. **Cause of anemia.**

   - Decreased serum iron (N=70-140 ug%)
   - Increased iron binding capacity (N=70%)
   - Reduced transferring saturation(N= 25-50%)
   - Reduced ferritin
   - BM examination: hyperpcellular with normoblasts

4. **Etiology:** e.g.

   - Anklystoma: Esinophilia, stool analysis for the worm or its ova
   - Occult blood in stool
   - Endoscopy
   - Platelet count may increase with active blood loss

5. **Iron treatment leads to reticulocytosis** (Therapeutic test)

NEW · ASH GUIDELINES 2026

Diagnosis of Iron Deficiency

What has changed in the new guidelines

Adults, menstruating & pregnant: serum ferritin cut-off

≤15 → ≤30 ng/mL

1. Cut-offs go up in every group studied
2. Inflammation: ferritin + TSAT together
3. Labs: clinical decision limits

Source: ASH 2026 guidelines for diagnosis of iron deficiency. Blood Adv. 2026; ASH Implementation & Quality Metric tools

New Ferritin Thresholds

NEW · 2026

Population	Old cut-off	New threshold	Consider a higher cut-off when…
Children 9 months – 4 years	≤12 ng/mL	**≤20 ng/mL**	Not specified (excludes inflammation, menstruation, pregnancy)
General adults Not menstruating or pregnant	≤15 ng/mL	**≤30 ng/mL**	≤50 with symptoms of iron deficiency or ongoing risk factors
Menstruating individuals	≤15 ng/mL	**≤30 ng/mL**	≤50 with heavy menstrual bleeding, symptoms, risk factors, upcoming surgery or planning pregnancy
Pregnant individuals	<15 ng/mL strong rec. against	**≤30 ng/mL**	<50 with anemia plus risk factors or symptoms

Why higher? Sensitivity of ferritin cut-offs in adults (19 studies)

41%	at ≤15 ng/mL
67%	at ≤30 ng/mL
89%	at ≤50 ng/mL

Source: ASH 2026 guidelines for diagnosis of iron deficiency. Blood Adv. 2026; ASH Implementation & Quality Metric tools

Inflammation & Lab Practice

NEW · 2026

Test both

In anemia of inflammation, order **serum ferritin + TSAT**, not ferritin alone. Sensitivity 38% with ferritin <100 alone vs 91% combined.

When to suspect

Ferritin <100 ng/mL or TSAT <20%

A guide, not a formal threshold (low-certainty evidence: heart failure, CKD, IBD).

Draw TSAT fasting

Most accurate after a **5–9 hour fast**. Recent iron-rich meals or supplements raise serum iron. CRP helps interpret, but is not a standalone test.

Labs should move from population reference ranges to **clinical decision limits**, so ferritin ≤30 ng/mL is flagged low wherever the patient is tested.

Source: ASH 2026 guidelines for diagnosis of iron deficiency. Blood Adv. 2026; ASH Implementation & Quality Metric tools

Treatment

I. Iron replacement:

1. Oral iron ferrous gluconate, ferrous sulphate.

- Complications: black stool(DD melena), constipation, nausea, vomiting and gastric ulcers.
- 4-10 weeks till HB is normal, then small doses for 3-6 months to fill stores
  - Hb stats to rise after one week and rises by 1gm/week
  - Reticulocyte count starts to increase on 4th day and persists for 12 days

2. Parenteral iron: iron dextran (IM, IV), iron sorbitol (IM)

- Indications: when oral therapy can not be used as in:
  - Peptic ulcer - Malabsorption - Complications of oral form
  - Need for rapid correction as in late pregnancy. - Failure of oral iron treatment.
- Complications: allergy, anaphylactic shock, thrombophelibitis, and hemochromatosis.

II. Treatment of cause: e.g.

- Anklystoma: Mebendazole (Vermox) : 100 mg BID for 3 days
- Treatment of peptic ulcer

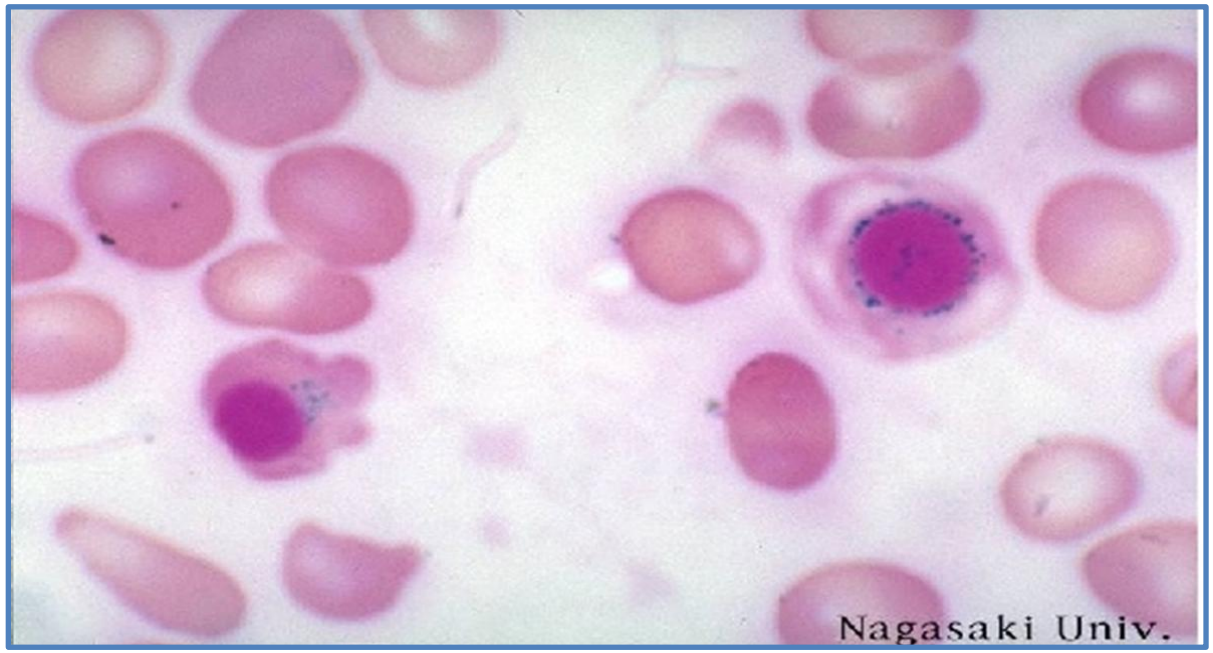
III. Blood transfusion if:

- HB < 7gm%
- Ischemic heart disease or heart failure if HB < 10 gm%

Sideroblastic anemia

Pathophysiology:

- Failure of incorporation of iron in heme synthesis due to defect in gamma amino-levulinic acid synthetase (ALA).
- The utilized iron will be deposited as hemsiderin in the mitochondria with formation of ringed sideroblasts. These cells are destroyed in the BM but some may appear in the peripheral blood.





Anemia in chronic diseases

Etiology

1. Chronic infections e.g. T.B
2. Chronic inflammation e.g. rheumatoid arthritis.
3. Malignancy

Pathogenesis

1. ↓RBCs survival
2. Reduced EPF response to anemia
3. Decreased iron release from stores to the developing erythroblasts

Clinical Picture As iron deficiency anemia

Investigations As iron deficiency anemia but:

1. There is reduced TIBC
2. BM: excess stainable sideroblasts
3. Raised acute phase reactants

Treatment Of the cause

DD of microcytic hypochromic anemia

	Iron deficiency	Anemia of chronic disease	Sideroblastic anemia	Thalassemia / Hb-pathies
Mechanism	↓ Iron → ↓ heme	Iron trapped (↑ hepcidin) + ↓ EPO	↓ Porphyrin (heme) synthesis	↓ Globin chain synthesis
RED CELLS				
MCV / MCHC	N → ↓	N → ↓	N → ↓	N → ↓
Retic count	N → ↓	N → ↓	N	↑
Smear	Aniso-/poikilocytosis, target cells	Usually nonspecific	Pappenheimer bodies (ring sideroblasts in marrow)	Many target cells (more than IDA)
IRON STUDIES				
Serum iron	↓	↓	↑	N → ↑
TIBC	↑	N → ↓	N → ↓	N → ↓
% Saturation	↓	↓	↑↑	N
Ferritin	↓	N → ↑	↑	N → ↑
Hepcidin	↓	↑	↓	↓
KEY CLUE	**Ferritin ↓ + TIBC ↑ (ASH 2026: ferritin ≤30)**	**Ferritin N/↑ + TIBC ↓ (hepcidin ↑)**	**Iron & saturation very ↑, ring sideroblasts**	**Iron normal, target cells → Hb electrophoresis**

Student pearl

The causes of microcytic anemia can be remembered with **IRON LAST**: **IRON** deficiency, Lead poisoning, Anemia of chronic disease, Sideroblastic anemia, Thalassemia.

Clinical pearl

Both iron deficiency anemia and anemia of chronic disease can manifest with normocytic anemia in the initial phase and microcytic anemia later on.

Clinical pearl

Bone marrow failure (e.g., due to myeloproliferative malignancy, myelodysplastic syndrome) can manifest with microcytic, normocytic, or macrocytic anemia.

Clinical pearl

Iron deficiency anemia and thalassemia trait are the most common causes of microcytic anemia. 8

Clinical pearl

Basophilic stippling on peripheral blood smear suggests lead poisoning or sideroblastic anemia. Because ringed sideroblasts are not usually seen in lead poisoning, they can help to distinguish between this condition and sideroblastic anemia.

Caution

While decreased ferritin confirms the diagnosis of iron deficiency anemia, elevated serum ferritin does not rule it out.

Hemorrhagic anemia

1. **Chronic hemorrhage: C/P as iron deficiency anemia (microcytic hypochromic).**
2. **Acute Hemorrhage (Loss of more than 1/L of blood acutely):**

- Immediately: There's proportional loss of RBCs & plasma → PCV is normal

  but with ↓ platelets & neutrophils within few hours
- Within 1 day: Redistribution of fluid from interstitial tissue to vascular space → hemodilution→ ↓PCV.
- After 3-4 days: ↑ BM activity → Reticulocytosis.

Clinical pearl

Hemoglobin and hematocrit levels can initially be normal in acute hemorrhage, even if there has already been significant blood loss. They will eventually decrease after plasma volume has been restored either spontaneously or via IV fluid resuscitation.

A 78-year-old man comes to the physician because of progressively worsening fatigue and headache for 6 months and occasional upper abdominal pain. He has not had any changes in his weight. He has a history of hypertension, type 2 diabetes mellitus, and chronic shoulder pain. Current daily medications include hydrochlorothiazide, metformin, and ibuprofen. He does not drink alcohol and does not smoke. His diet has mostly consisted of potatoes and eggs since his wife passed away 1 year ago. His pulse is 92/min and blood pressure is 135/80 mm Hg. Examination shows conjunctival and mucosal pallor. Abdominal and rectal examination shows no abnormalities. Neurologic examination shows mild postural tremor. Peripheral blood smear shows red blood cells with increased zones of central pallor and anisocytosis.

Which of the following is the most likely underlying cause of this patient's symptoms?

a) Lead poisoning

b) Folic acid deficiency

c) Bone marrow failure

d) Intravascular hemolysis

e) Impaired erythropoietin production

f) Vitamin B12 deficiency

g) Iron deficiency

Summary



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Thank You